

CARLOFORTE, Italy

WMO: 165490

Lat: 39.137N

Lon: 8.312E

Elev: 39

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 86-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 42.4	(c) 44.6	(d) 37.0	(e) 32.4	(f) 67.0	(g) 38.4	(h) 34.2	(i) 71.0	(j) 31.0	(k) 56.1	(l) 27.7	(m) 54.7	(n) 7.6	(o) 70	(p) 0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 15.1	(c) 89.5	(d) 64.5	(e) 87.4	(f) 64.8	(g) 85.4	(h) 65.0	(i) 76.3	(j) 81.7	(k) 74.6	(l) 79.6	(m) 73.1	(n) 77.9	(o) 8.0	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	130.2	79.0	73.0	122.7	78.1	71.3	115.7	76.7	39.8	81.3	38.2	79.9	36.7	78.2	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.7	(b) 23.2	(c) 20.5	(e) 39.6	(f) 95.1	(g) 2.7	(h) 1.9	(i) 37.6	(j) 96.5	(k) 36.0	(l) 97.6	(m) 34.5	(n) 98.7	(o) 32.5	(p) 100.1		
(4) 26.7	(5) 23.2	(6) 20.5	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.7	53.4	54.4	56.4	59.1	66.2	72.3	77.7	78.5	73.6	68.4	60.6	55.3
	DBStd	9.72	3.55	3.28	3.63	3.60	4.27	4.27	3.44	3.00	4.24	4.70	4.02	3.90
	HDD50	30	14	6	3	0	0	0	0	0	0	0	0	7
	HDD65	1605	360	296	268	180	36	3	0	0	1	21	138	302
	CDD50	5402	119	130	200	272	503	669	857	885	708	572	317	170
	CDD65	1503	0	0	0	2	74	222	392	420	260	128	5	0
	CDH74	9465	0	0	0	7	239	1235	3049	3273	1320	341	1	0
	CDH80	2580	0	0	0	1	16	241	989	1063	250	20	0	0
Wind	WSAvg	8.3	8.1	7.7	8.6	9.7	7.6	7.3	7.6	7.8	7.6	8.1	9.2	9.7
Precipitation	PrecAvg	22.4	2.3	2.2	1.9	2.2	1.3	0.6	0.2	0.4	1.4	2.8	3.8	3.3
	PrecMax	31.7	6.0	4.7	3.9	4.9	3.0	2.6	0.6	1.7	4.4	6.6	9.3	6.1
	PrecMin	15.0	0.7	0.4	0.3	0.5	0.2	0.1	0.1	0.0	0.1	0.6	0.9	1.0
	PrecStd	4.2	1.3	1.2	0.9	1.1	0.7	0.6	0.2	0.4	1.0	1.4	1.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.7	65.8	69.4	72.7	82.2	87.9	93.3	92.2	87.8	82.5	73.0	67.4
		MCWB	58.2	54.6	55.9	56.1	60.5	61.9	62.9	63.6	66.7	62.1	64.8	62.2
	2%	DB	61.5	63.6	66.6	69.8	78.9	85.2	89.8	89.8	84.9	79.5	70.4	64.6
		MCWB	56.2	54.7	55.0	57.2	59.4	62.8	65.7	63.7	65.3	63.5	61.2	57.4
	5%	DB	60.1	61.8	64.7	67.9	76.8	82.8	87.5	87.6	82.9	77.7	68.8	62.7
		MCWB	55.2	54.5	54.6	56.7	59.7	62.0	66.1	63.8	65.3	64.1	59.6	57.1
	10%	DB	58.8	60.3	62.8	65.8	74.3	80.6	85.4	85.6	80.9	75.7	66.7	60.9
		MCWB	54.6	53.9	54.2	56.1	58.6	60.4	66.5	64.3	64.7	64.4	58.6	55.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.4	60.4	61.2	63.4	68.8	74.7	78.7	77.3	76.7	72.5	67.7	63.5
		MCDB	63.6	62.0	64.6	67.0	74.4	80.8	83.3	83.5	80.5	75.9	71.2	65.1
	2%	WB	59.1	58.5	59.2	61.8	66.1	72.7	77.0	75.3	74.6	70.8	65.2	61.2
		MCDB	60.6	60.4	62.4	65.6	72.1	77.0	82.7	80.5	79.0	74.3	68.7	62.6
	5%	WB	57.8	57.0	57.3	60.6	64.6	68.0	75.2	73.7	72.6	69.2	63.2	59.2
		MCDB	59.4	59.1	59.3	64.3	69.9	76.2	80.2	78.9	77.3	72.5	66.4	60.8
	10%	WB	56.2	55.2	55.6	59.1	63.0	63.0	73.4	72.1	70.2	67.5	61.0	56.9
		MCDB	57.9	57.1	58.5	62.8	68.6	75.5	78.4	76.8	74.4	70.5	63.9	59.0
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.8	12.0	12.4	14.4	15.2	15.9	15.1	13.3	11.3	10.3	9.7
		MCDBR	11.6	13.3	14.5	15.4	17.1	16.8	18.4	17.3	14.5	12.8	12.5	10.9
		MCWBR	6.8	6.3	5.6	5.7	5.4	5.3	6.4	6.0	5.4	5.7	6.2	6.0
	5% WB	MCDBR	10.4	11.2	11.7	12.3	13.2	13.7	14.8	15.2	13.5	11.7	12.1	10.3
		MCWBR	8.6	8.4	8.0	7.6	7.6	9.4	8.9	9.2	8.2	7.3	8.1	8.4
Clear-Sky Solar Irradiance	taub	0.344	0.364	0.402	0.438	0.453	0.432	0.448	0.437	0.433	0.409	0.380	0.353	
	taud	2.468	2.400	2.290	2.205	2.199	2.306	2.265	2.305	2.307	2.345	2.375	2.432	
	Ebn at Noon	261	270	270	267	264	269	264	264	257	250	243	247	
	Edh at Noon	26	31	38	44	46	41	43	40	37	33	28	25	
All-Sky Solar Radiation	RadAvg	663	950	1391	1772	2110	2426	2434	2144	1617	1162	737	597	
	RadStd	55	90	126	115	133	75	76	91	61	81	51	55	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page